SECTION A [25 *marks*]

This section consists of **3** questions. Answer **all** questions.

1 Solve

(a) $\lim\limits_{x \to -2} \dfrac{x^3+8}{x^2-4}$

[4 *marks*]

(b) $\lim\limits_{x \to 2} \dfrac{x-2}{3-\sqrt{x^2+5}}$

[5 *marks*]

2 By expressing $y=\dfrac{x}{(x+2)^2}$ into partial fraction, find $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$. Give your answer in the simplest form. Hence, evaluate $\dfrac{d^2y}{dx^2}$ when $x = 0$.

[9 *marks*]

3 Find the critical numbers for the curve $y=3x^4-4x^3$. Hence, determine the points of inflection.

[7 *marks*]

SECTION B [75 *marks*]

This section consists of **7** questions. Answer **all** question.

1 A complex number z satisfy the equation $\dfrac{z}{z+2} = 2 - i$.

State the conjugate of z.

[5 *marks*]

2 Solve

(a) $5 \times 5^{\log x} + 5^{2-\log x} = 30$.

[5 *marks*]

(b) $2 \leq |5 - x| < 7$.

[6 *marks*]

3 The functions f and g are defined by

$$f(x) = x^2 + 2kx + 4, \;\; x \in R \text{ and}$$

$$g(x) = 3 - kx, \;\; x \in R$$

where k is a non zero constant.

(a) Find the range of $f(x)$ in terms of k.

[2 *marks*]

(b) If $(f \circ g)(2) = 4$, show that $k = \dfrac{3}{2}$.

[2 *marks*]

(c) Given $(h \circ g)(x) = \dfrac{2}{8-3x}$, determine the function of $h(x)$.

[3 *marks*]

4 (a) Given $f(x)=\frac{1}{2}\left(e^{4-x}+3\right)$ and $g(x)=4-\ln(2x-3)$.

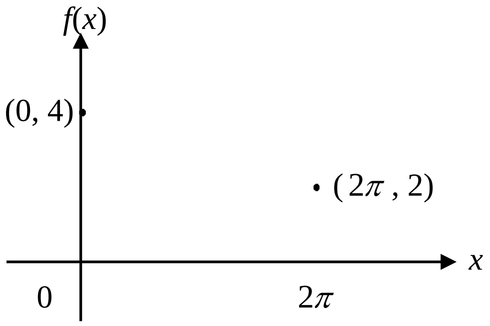
(i) Find $(f \circ g)(x)$ and $(g \circ f)(x)$. Hence, state $f^{-1}(x)$ and its domain and range.

[6 *marks*]

(ii) Sketch the graphs of $g(x)$ and $g^{-1}(x)$ on the same axes.

[3 *marks*]

(b)



The graph above shows the function $f(x)=m+\cos nx,\ 0\le x\le 2\pi$.

State the range of *f*(*x*). Hence, find the values of *m* and *n*.

[4 *marks*]

5 (a) Evaluate $\lim_{x\to-\infty}\frac{\sqrt{x+4x^2}}{3x-1}$

[4 *marks*]

(b) Given $f(x)=\begin{cases}\frac{|x-2|}{x-2} & , \quad x<2\\ ax^2-bx+3 & , \quad 2\le x<3\\ 2x-a+b & , \quad x\ge 3\end{cases}$.

Find *a* and *b* such that the function is continuous for all *x*.

[5 *marks*]

(c) The curve *C* has equation $y=f(x)$ given by $f(x)=\frac{x^2}{(x-1)(x-5)}$. By finding the limit at appropriate points, obtain the horizontal and vertical asymptotes.

[5 *marks*]

6 (a) Given $y^4 = e^{x-y}$. Show that

(i) $\dfrac{dy}{dx} = \dfrac{y}{4+y}$.

[4 *marks*]

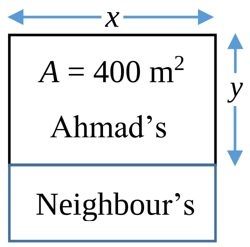
(ii) $(4+y)^3 \dfrac{d^2y}{dx^2} - 4y = 0$.

[4 *marks*]

(b) Given $x = \sqrt{\sin 2t}$ and $y = \sqrt{\cos 2t}$. Find $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$. Give your answer in the simplest form.

[8 *marks*]

7 Ahmad wants to build a fence to protect a rectangular 400 m^2 planting area. His next-door neighbour agrees to pay for half of the fence that borders his property. Ahmad will pay for the rest of the cost. What are the dimensions of the planting area in order for the total cost of building the fence to be a minimum?



[9 marks]

END OF QUESTION PAPER

Answers

Section A

1. (a) -3
   (b) $-\dfrac{3}{2}$

2. $\dfrac{1}{x+2}-\dfrac{2}{(x+2)^2}$; $-\dfrac{1}{(x+2)^2}+\dfrac{4}{(x+2)^3}$; $\dfrac{2}{(x+2)^3}-\dfrac{12}{(x+2)^4}$; $-\dfrac{1}{2}$

3. $x=0,1$; $(0,0)$

Section B

1. $-3-i$, -2.82

2. (a) $x=1,10$
   (b) $(-2,3]\cup[7,12)$

3. (a) $[4-k^2,\infty)$
   (b) $\dfrac{3}{2}$
   (c) $\dfrac{1}{1+x}$

4. (a) i. x , $4-\ln(2x-3)$, $D_{f^{-1}}=\left(\dfrac{3}{2},\infty\right)$, $R_{f^{-1}}=(-\infty,\infty)$

   ii. DIY

   (b) $m=3$, $n=\dfrac{1}{2}$

5. (a) $-\dfrac{2}{3}$
   (b) $a=\dfrac{11}{2}$, $b=13$
   (c) $x=1$ or $x=5$ VAs , $y=1$ HA

6. (a) i. DIY
   ii. DIY
   (b) $-\tan^{\frac{3}{2}}2t$, $3\tan 2t\sec^2 2t$

7. $x=\dfrac{40}{\sqrt{3}}$ m and $y=10\sqrt{3}$ m